

Marika Nishi Mobile

Mobile Robots Intern - Mechatronics Vehicle Feature(s) R&D
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EDUCATION

University of Pennsylvania, School of Engineering & Applied Science | Philadelphia, PA, United States May 2027
Candidate for Master of Science in Engineering, *Majors: Robotics, Computer Science* | *Concentration: Computer Vision* GPA: 3.78/4.0
University of Tokyo, Faculty of Engineering | Bunkyo, Tokyo, Japan March 2023
Bachelor of Engineering, *Major: Mechanical Engineering* GPA: 3.64/4.0

SKILLS & RELEVANT COURSEWORK

Software Skills: C++, C, Python, PyTorch, Robot Operating System (ROS), motion planning, Simulator (Gazebo), mobile robots, perception, 1/10th scale cars, ML, ML-based planning, CUDA, control, segmentation, SLAM, Sensor Fusion, OpenCV

Hardware Skills: LiDAR, RGB cameras, GPS, IMU, EKF, actuators, 3D printers, laser cutters, drill presses, microcontroller

Relevant Coursework: Mechatronics; F1tenth Autonomous Racing Cars; Automotive Engineering; Deep Neural Networks; Deep Learning; Computer Vision; Dynamics and Control; Machine Perception; Machine Learning

Publication: Hori, K., Watanabe, K., Tanaka, T., **Nishi, M.**, & Ito, T. (2026). Visibility Evaluation Around Unsignalized Intersections with Occlusion From Arbitrary On-road Viewpoints Using Point Cloud Maps. *International Journal of Intelligent Transportation Systems Research*.

Watanabe, K., **Nishi, M.**, & Ito, T. (2025). Placement design of various roadside sensors for stochastic position prediction of traffic participants on community roads. *International Journal of Intelligent Transportation Systems Research*.

EMPLOYMENT EXPERIENCE

Vijay Kumar Lab | *Research Intern*, Philadelphia, PA, United States May 2025 – Present

- Implement driving simulators that visualized data of event camera on F1tenth car using Gazebo, CARLA, and ROS
- Collect datasets on driving simulator and train vision transformer model that computed appropriate motion control from event camera images; improve data generation process efficiency by removing redundancy, cutting time by 70 %

BOSCH | *Software Intern*, Tsuzuki, Kanagawa, Japan August 2023 – September 2023

- Designed and built map to enhance autonomous driving by visualizing key information, such as routes and vehicle state using MATLAB; helped company's engineers understand signals sent from cars visually
- Improved software that processed CAN signals from cars by fixing signal decoding; refined system output accuracy
- Collaborated with colleagues from 10 countries and different majors; made outcomes reflecting diverse opinions

DMG MORI | *Industrial Practices Intern, Additive Manufacturing R & D*, Iga, Mie, Japan August 2021 – September 2021

- Analyzed performance tests of additive manufacturing machines by operating the machines and inspecting products
- Investigated additive manufacturing methods by examining machine structures; proposed solutions to heat distortion

ROBOT AUTOMATION PROJECTS

Mobile Robots Competition | *Mechatronics, CAD, Mechanical Design, Microcontroller, Solder, Control, Webpage, C* January 2026 – Present

- Design and fabricate a mobile robot using CAD, laser cutting, 3D printing
- Develop embedded control software in C (Arduino) for real-time motor control and sensor integration
- Implement wall-following using ToF sensor feedback and closed-loop control
- Design and solder custom circuitry and build a web-based interface for wireless robot control

Research of Traffic Collision Prediction System | *Prototyping, Extended Kalman Filter, C++, ROS* April 2022 – March 2023

- Designed and built real-time collision prediction system among pedestrian, cyclist, and intelligent wheelchair
- Wrote C++ ROS codes that detected pedestrians and cyclists based on LiDAR data
- Conducted sensor fusion using Extended Kalman Filter; estimated pedestrians and cyclists' positions
- Operated RTK-GPS, low-accuracy GPS, LiDAR, and IMU to collect and integrate data into system development